

For loops - answer key

Instructor copy | Accept equivalent correct explanations and code.

1. Trace the output

2, 5, 8. The body runs three times. After the third update, i is 11, so $i < 9$ is false.

Ask students to distinguish a body execution from a condition check.

2. Check the boundary

Six messages are printed, for $i = 0$ through 5. Change the condition to $i < 5$ (or $i \leq 4$) to print five.

The initialization remains $i = 0$; only the condition changes.

3. Can the body be skipped?

Nothing is printed. The first condition check, $7 < 4$, is false. Neither the body nor the update executes.

The update does not happen before the first check.

4. Write a loop

```
for (int n = 12; n >= 0; n -= 3) {  
    println(n);  
}
```

Accept equivalent loops that print exactly 12, 9, 6, 3, 0. The next value is -3, which fails the condition.

5. A loop inside draw()

Four circles appear in each completed frame, at $x = 40, 80, 120$, and 160 . The entire loop runs during each call to `draw()`; its iterations are not separate frames.

The background clears the previous frame. Each new call initializes the loop variable to 40 again.